Programming Assignment Report

Design and Analysis of Algorithms

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Problem Name: Task Scheduling

1. Problem Statement

Task Scheduling involves completing tasks where each task has a deadline (Di) and a time required (Mi). The goal is to schedule tasks such that the maximum delay beyond deadlines is minimized.

2. Hackerrank

https://www.hackerrank.com/challenges/task-scheduling/problem

3. Github Submission Link

https://github.com/SHREYADAA/DAA-Assignment

4. Algorithm

Start the program.

Input number of tasks (T).

For each task, input deadline (Di) and time (Mi).

Store tasks in a list.

Sort tasks based on deadlines.

Keep track of total completion time.

For each task:

- Add its time to total time.
- Calculate overshoot = total time − deadline.

- Track maximum overshoot.

Print the maximum overshoot for each step.

Stop the program.

5. Example Explanation

Input:
5
2 2
1 1
4 3
10 1
2 1

Output:
0
1
2
2
3

6. Complete Java Code

```
import java.util.*;

class Task {
   int deadline, time;

   Task(int d, int t) {
      deadline = d;
      time = t;
   }
}

public class TaskScheduling {

   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      int T = sc.nextInt();
      List<Task> tasks = new ArrayList<>();
```

```
        for (int i = 0; i < T; i++) {
            int d = sc.nextInt();
            int m = sc.nextInt();
            tasks.add(new Task(d, m));
        }

        tasks.sort(Comparator.comparingInt(t -> t.deadline));

        int totalTime = 0;
        int maxOvershoot = 0;

        for (int i = 0; i < T; i++) {
            totalTime += tasks.get(i).time;
            int overshoot = Math.max(0, totalTime - tasks.get(i).deadline);
            maxOvershoot = Math.max(maxOvershoot, overshoot);
            System.out.println(maxOvershoot);
        }

        sc.close();
    }
}
```

7. Observation / Solution

Tasks are scheduled based on deadlines.
Sorting ensures earlier deadlines are handled first.
Overshoot is calculated as delay beyond deadline.
The algorithm minimizes the maximum delay.

8. Time Complexity

Time Complexity: O(T log T)